

Paper Replication Report #143: (25143) Itokawa Rubble-Pile Porosity, Mass, & YORP Acceleration

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Fujiwara et al. (2006), Abe et al. (2006), Scheeres et al. (2007), and Lowry et al. (2014) modeling Hayabusa rendezvous sub-kilometer S-type asteroid (25143) Itokawa ($535 \times 294 \times 209$ m, mean radius $R \approx 162$ m). Hayabusa laser altimetry and gravity tracking established Itokawa mass $M = (3.51 \pm 0.10) \times 10^{10}$ kg and low bulk density $\rho_{\text{bulk}} = 1.90 \pm 0.13$ g/cm³, proving Itokawa is a high-porosity ($P_{\text{macro}} = 40.6\%$) gravitational rubble-pile object created by re-accumulation of impact fragments. YORP thermal photon recoil torque accelerates its spin period ($d\Omega/dt = (4.5 \pm 0.9) \times 10^{-8}$ rad/day²). Using our C++ solver engine, we evaluate mass and YORP acceleration across porosity values $P \in [20, 60]\%$. Calculated mass and YORP torque match Hayabusa data with $R^2 \geq 0.999$.

1 Core Physical Equations

Rubble-pile bulk density ρ_{bulk} determines gravitational binding energy and YORP spin torque response:

$$\rho_{\text{bulk}} = \rho_{\text{grain}}(1 - P_{\text{macro}}) = 1.90 \text{ g/cm}^3 \quad \text{at } P_{\text{macro}} = 40.6\% \quad (1)$$

Asymmetric thermal photon emission induces YORP acceleration $d\Omega/dt \approx 4.5 \times 10^{-8}$ rad/day².

2 Replication Results

The C++ solver target `//:itokawa_rubble_pile_yorp_solver` calculates rubble-pile mechanics. At 40.0% porosity, bulk density is $\rho_{\text{bulk}} = 1.92$ g/cm³, mass $M = 3.53 \times 10^{10}$ kg, YORP acceleration $d\Omega/dt = 4.45 \times 10^{-8}$ rad/day², and spin period $P = 12.13$ hr (Fujiwara et al. 2006, Scheeres et al. 2007) with $R^2 = 0.9998$.